

An explicit joint Laplace transform for Brownian first-visit cell lengths on a circle

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Abstract

We give a complete analytic-transform answer to Georgakopoulos's question about the lengths of the sets first visited by independent Brownian particles on a circle, with uniform random or equally spaced starting points. For every finite number of particles, the joint Laplace transform is an absolutely convergent series whose coefficients are finite sums of ordinary integrals of explicit interval-exit kernels. The integration regions are specified linear inequalities in cumulative physical hitting times. The outer series has a uniform factorial remainder bound and uniquely determines the probability measure. The result also applies to arbitrary fixed distinct starting points. The construction uses classical heat kernels, strong Markov iteration, and bounded-support moments. Earlier competing-walk painting results are credited and distinguished. No compact density, efficient numerical algorithm, or absolute historical priority is asserted.

1 The question and the model

Georgakopoulos's question on printed p. 1452 of Oberwolfach Report 23/2018 asks for the joint lengths of first-visit cells of k independent Brownian particles started uniformly at random or at equally spaced points of S^1 [1]. The report belongs to the 2018 volume and was published on April 12, 2019. We answer the finite- k distribution question by an explicit transform law; no named density or large- k limit is imposed in that question.

Normalize the circle to $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, with Lebesgue measure one. Let $k \geq 1$ and let W_i be independent Brownian motions with generator $\frac{1}{2}d^2/dx^2$, starting at distinct labeled points p_i . The particles continue moving independently through all encounters, seeds, and previously visited territory. Put

$$T_i(x) = \inf\{t \geq 0 : W_i(t) = x\}, \quad I(x) = \min_{1 \leq i \leq k} \arg \min T_i(x), \quad L_i = \int_{\mathbb{T}} \mathbf{1}_{\{I(x)=i\}} dx. \quad (1)$$

We justify measurability and the irrelevance of ties below. For circumference $c > 0$, the lengths are cL_i , so replace each Laplace variable θ_i by $c\theta_i$. A common positive diffusivity change rescales all clocks equally and leaves ownership unchanged.

2 Explicit deterministic coefficients

For $0 < \alpha < \ell$ and $t > 0$, define

$$f_0(t; \alpha, \ell) = \frac{\pi}{\ell^2} \sum_{n=1}^{\infty} n \sin \frac{n\pi\alpha}{\ell} e^{-n^2\pi^2 t/(2\ell^2)}, \quad (2)$$

$$f_1(t; \alpha, \ell) = \frac{\pi}{\ell^2} \sum_{n=1}^{\infty} (-1)^{n+1} n \sin \frac{n\pi\alpha}{\ell} e^{-n^2\pi^2 t/(2\ell^2)}. \quad (3)$$

These are the nonnegative densities of interval exit at 0 and ℓ , respectively. The series are absolutely convergent at each $t > 0$. They are summed before any improper integration near $t = 0$; no termwise integration of the signed Fourier series is required.

For a nonempty finite set $A \subset \mathbb{T}$ and $z \notin A$, lift the component of $\mathbb{T} \setminus A$ containing z to $(a, b) \subset \mathbb{R}$, and represent z by $\tilde{z} \in (a, b)$. Set $\ell = b - a$, $\alpha = \tilde{z} - a$, and, for $q \in A$, define

$$K_A(t; z, q) = \mathbf{1}_{\{a \bmod 1=q\}} f_0(t; \alpha, \ell) + \mathbf{1}_{\{b \bmod 1=q\}} f_1(t; \alpha, \ell). \quad (4)$$

Integer changes of lift do not change this formula. For a singleton $A = \{q\}$, both endpoints represent q , $\ell = 1$, and the kernel is $f_0 + f_1$. Previously visited targets are not absorbing.

Fix $m \geq 1$ distinct query points $x = (x_1, \dots, x_m)$, none a seed. Write S_m for the permutations of $\{1, \dots, m\}$. For $\pi \in S_m$ and particle i , let

$$A_r^\pi = \{x_{\pi(r)}, \dots, x_{\pi(m)}\}, \quad z_{i,0} = p_i, \quad z_{i,r-1} = x_{\pi(r-1)} \quad (r \geq 2).$$

For $u_i = (u_{i1}, \dots, u_{im}) \in (0, \infty)^m$, set

$$D_i^\pi(u_i; x, p_i) = \prod_{r=1}^m K_{A_r^\pi}(u_{ir}; z_{i,r-1}, x_{\pi(r)}). \quad (5)$$

For $\boldsymbol{\pi} = (\pi_1, \dots, \pi_k) \in S_m^k$, reconstruct the physical hitting times, rather than comparing the separate increments:

$$h_{ij}(u, \boldsymbol{\pi}) = \sum_{r=1}^{\pi_i^{-1}(j)} u_{ir}. \quad (6)$$

For a label vector $a = (a_1, \dots, a_m) \in \{1, \dots, k\}^m$, put

$$\mathcal{C}_a(\boldsymbol{\pi}) = \{u > 0 : h_{a_j,j}(u, \boldsymbol{\pi}) < h_{ij}(u, \boldsymbol{\pi}) \text{ for every } j \text{ and } i \neq a_j\}, \quad (7)$$

$$Q_a(x; p) = \sum_{\boldsymbol{\pi} \in S_m^k} \int_{\mathcal{C}_a(\boldsymbol{\pi})} \prod_{i=1}^k D_i^{\pi_i}(u_i; x, p_i) \prod_{i=1}^k \prod_{r=1}^m du_{ir}. \quad (8)$$

Thus Q_a is specified by $(m!)^k$ nonnegative, km -dimensional integrals of the explicit scalar series, over strict linear inequalities with integer coefficients. Impossible orders have zero factors.

For $\theta_i \geq 0$, let $\theta_* = \max_i \theta_i$, define $M_0(\theta; p) = 1$, and put

$$M_m(\theta; p) = \int_{\mathbb{T}^m} \sum_{a \in \{1, \dots, k\}^m} \left(\prod_{j=1}^m \theta_{a_j} \right) Q_a(x; p) dx_1 \cdots dx_m \quad (m \geq 1). \quad (9)$$

Assign arbitrary bounded values on the excluded query configurations; they have zero Lebesgue measure. All coefficient definitions are deterministic. In particular, they contain no unknown Brownian hitting probability, conditional law, or integral-equation solution.

Theorem 1. For every finite $k \geq 1$ and every distinct seed vector p , $L = (L_1, \dots, L_k)$ is well defined almost surely, lies in the simplex $\{l_i \geq 0 : \sum_i l_i = 1\}$, and has the joint Laplace transform

$$\mathcal{L}_p(\theta) = \mathbb{E}_p e^{-\sum_i \theta_i L_i} = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} M_m(\theta; p). \quad (10)$$

The series is absolutely convergent, $0 \leq M_m \leq \theta_*^m$, and, for every $N \geq 0$,

$$\left| \mathcal{L}_p(\theta) - \sum_{m=0}^N \frac{(-1)^m}{m!} M_m(\theta; p) \right| \leq \frac{\theta_*^{N+1}}{(N+1)!}. \quad (11)$$

The coefficients uniquely determine the entire joint probability measure. For equally spaced seeds use $p_i = (i-1)/k$. For independent uniform seeds, the answer is

$$\mathcal{L}_{\text{unif}}(\theta) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} \int_{\mathbb{T}^k} M_m(\theta; p) dp, \quad (12)$$

with the same bound (11).

3 Proof

Lemma 1. The kernels in (2)–(4) are the next-target exit-time/place densities, and

$$\int_0^\infty f_0 dt = 1 - \alpha/\ell, \quad \int_0^\infty f_1 dt = \alpha/\ell, \quad \sum_{q \in A} \int_0^\infty K_A(t; z, q) dt = 1. \quad (13)$$

Proof. The classical Dirichlet heat kernel, obtained from the interval sine basis and space–time scaling, is

$$H(t; \alpha, \beta) = \frac{2}{\ell} \sum_{n \geq 1} \sin \frac{n\pi\alpha}{\ell} \sin \frac{n\pi\beta}{\ell} e^{-n^2\pi^2 t / (2\ell^2)}. \quad (14)$$

See also Lalley’s Exercise 6(C) [2] for the unit interval. If τ is the interval exit time, stopping $B_t^2 - t$ at $\tau \wedge t$ bounds $\mathbb{E}(\tau \wedge t)$ uniformly, since the stopped position stays in $[0, \ell]$. Hence $\tau < \infty$ almost surely. Optional stopping for the bounded stopped coordinate gives exit probabilities $1 - \alpha/\ell$ and α/ℓ .

The Markov property therefore gives

$$\mathbb{P}_\alpha(\tau > t, B_\tau = 0) = \int_0^\ell H(t; \alpha, \beta)(1 - \beta/\ell) d\beta.$$

For $t > 0$, locally uniform convergence permits differentiation. Use $H_t = H_{\beta\beta}/2$, integrate twice by parts, and use the Dirichlet boundary values. The negative derivative is $\frac{1}{2}H_\beta(t; \alpha, 0) = f_0(t; \alpha, \ell)$. With weight β/ℓ it is $-\frac{1}{2}H_\beta(t; \alpha, \ell) = f_1(t; \alpha, \ell)$. This proves nonnegativity, the density interpretation, and the first two identities in (13), without improper Fourier interchanges. Circle continuity identifies the next target with the exit endpoint of its lifted component. Summing both endpoints, including their identification for a singleton, proves the last identity. $\square$

For comparison, the familiar interval-exit Laplace transforms are

$$\int_0^\infty e^{-st}(f_0, f_1) dt = \left(\frac{\sinh((\ell - \alpha)\sqrt{2s})}{\sinh(\ell\sqrt{2s})}, \frac{\sinh(\alpha\sqrt{2s})}{\sinh(\ell\sqrt{2s})} \right) \quad (s > 0). \quad (15)$$

These follow from $v''/2 = sv$ with the corresponding endpoint data; see, for example, Ernst–Shepp [4]. They are normalization checks, not new interval formulas.

Lemma 2. D_i^π is the joint density of the order π and elapsed increments between successive newly visited queries for particle i . Moreover,

$$\sum_{\pi \in S_m} \int_{(0, \infty)^m} D_i^\pi(u_i) du_i = 1, \quad Q_a(x; p) = \mathbb{P}_p(I(x_1) = a_1, \dots, I(x_m) = a_m), \quad \sum_a Q_a(x; p) = 1.$$

Proof. Stop a single walker when it first reaches the finite query set, remove that reached target, and repeat with the remaining set. At each step the current point lies outside the remaining targets. These are finite stopping times by Lemma 1. Distinct queries cannot be reached simultaneously by a continuous path. The strong Markov property [3, Theorem 2.16] gives exactly the product (5); repeated normalization in (13) gives its total mass. This argument tracks the same physical walker, which may cross already visited queries and other particles' territory.

Independence gives the product of the k ordered densities. Equation (6) reconstructs each physical first-hitting time, so the owner event is exactly (7). A fixed query's individual hitting-time distribution is atomless by the singleton kernel. Independent walkers thus have no tie at a fixed query, almost surely. The finite union over queries is still null; equivalently the equality boundaries are proper hyperplanes of zero measure for these densities. Integrating and summing all orders proves the assertions about Q_a . $\square$

Proof of Theorem 1. On continuous-path space, the visited set up to a fixed time t is compact. Its distance from x is the infimum of the distances from positions at rational times in $[0, t]$, together with the position at t . Hence $(\text{path}, x) \mapsto T_i(x)$ is jointly measurable. For each nonseed x , every particle reaches it in finite time and the independent hitting times are atomless. Fubini shows that, almost surely, locations with any infinite hitting time or an earliest tie have measure zero. Thus (1) defines a measurable partition up to a null set and $\sum_i L_i = 1$. This is not an assertion that ties are absent simultaneously at every location.

Put $Z = \sum_i \theta_i L_i = \int_{\mathbb{T}} \theta_{I(x)} dx$, so $0 \leq Z \leq \theta_*$. Tonelli and Lemma 2 yield $\mathbb{E}_p Z^m = M_m(\theta; p)$ and $0 \leq M_m \leq \theta_*^m$. The exponential series is absolutely dominated by e^{θ_*} ; termwise expectation proves (10). Taylor's theorem for e^{-z} on $z \geq 0$ bounds the degree- N remainder by $z^{N+1}/(N+1)!$, giving (11).

The homogeneous moment polynomials give all mixed moments, or these can be recovered by differentiation of the transform at zero, justified by bounded support. Polynomials are dense in continuous functions on the compact simplex by Stone–Weierstrass. Consequently its finite Borel measures are uniquely determined by these moments. The remainder and absolute bounds are uniform in distinct p . Coincident independent-uniform seeds have measure zero, so integration in p and interchange with the series prove (12). Equally spaced starts are already a fixed- p instance. $\square$

A random rotation of equally spaced seeds leaves lengths unchanged. If seeds instead receive labels after circular ordering, integrate the same fixed- p transform against that ordered seed law. Formula (12) uses independent uniform labels. For $k = 1$ the transform is $e^{-\theta_1}$. If every $\theta_i = t$, then $M_m = t^m$ and the transform is e^{-t} . Uniform starts are exchangeable; equally spaced starts have cyclic and dihedral symmetry. Both give $\mathbb{E} L_i = 1/k$; full permutation exchangeability for equally spaced labels is not asserted.

4 Attribution, priority and verification

First-arrival painting is an established model. Gomes Júnior et al. [5] treat a two-walker periodic lattice and a one-point scaling probability. Miller [6] studies ownership correlations and asymptotic volume variance on transient graph families. Chatterjee–Nabahi–Terlov [7] analyze expected

interfaces on a discrete cycle. Baccara’s joint real-line competitive-range limit [8, Section 6, Proposition 8] remains a Brownian functional characterization; Si [9] studies strategic painting payoffs, including a reflected-Brownian interval limit. These are relevant predecessors with different observables or regimes.

Moment-based transforms and exit-increment constructions are classical; see Kac’s product-moment framework as reviewed by Fitzsimmons–Pitman [11] and the multitime single-walker span laws of Régnier et al. [12]. The reduction here makes every finite-query ownership probability explicit while preserving each walker’s correlated first-hitting times, then determines the joint circle volume law. No novelty is claimed for the model, interval kernels, strong Markov property, moment expansion, or moment determinacy.

Our bounded primary-source audit of October 7, 2026 did not identify an earlier complete characterization of this finite- k circle law, or a general explicit result directly supplying all its coefficients. The full Gomes 1996 article and Dicker’s related higher-dimensional thesis [10] were inaccessible; only the former’s publisher abstract and later primary discussions, and the latter’s primary index, were available. Direct Zenodo registry searches also failed. These limitations, search records, and theorem-scope comparisons are retained in the supplement. No absolute priority, exhaustive absence, or independent-discovery claim is made.

The supplied diagnostics reproduce exact finite-chain order, clock, and moment identities and compare scalar exit fluxes by independent spectral and reflected-Gaussian formulas. They test supporting calculations; they are not a Brownian limit proof or a certificate of numerical quadrature. The proof above establishes the Brownian result. The outer bound (11) does not bound numerical errors in the coefficient integrals or a truncation of their inner kernel series. A compact density, named distribution, and efficient evaluator remain possible refinements.

AI and review disclosure. AI tools were used extensively in developing, drafting, reproducing, and adversarially verifying this work. Independent AI reviews are not conventional human peer review. This preprint is unrefereed and has not undergone conventional human peer review.

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